

HDOC Day-40

Menu-Driven Student Name Operations Using Strings and Pointers in C

Algorithm

1. Read the student's first name, middle name, and last name.
2. Combine the three names using strcpy() and strcat() to create fullName.
3. Display a menu repeatedly until the user selects Exit.
4. Choice 1: combine and display the full name.
5. Choice 2: obtain strlen(fullName) through findLength() and return the length through a pointer.
6. Choice 3: copy fullName to another string using strcpy().
7. Choice 4: compare firstName and lastName using strcmp() and return the comparison result through a pointer.
8. Choice 5: form reverseName as last name, middle name, first name.
9. Choice 6: scan fullName and count vowels, consonants, digits, and spaces; return all four counts through pointers.
10. Choice 7: read a word and use strstr() to determine whether it occurs in fullName; return the result through a pointer.
11. Choice 8: terminate the program. For any other choice, display an error message.

Test Case Table

TC	Choice	Input	Expected Result
1	1	First=Mahak, Middle=Kumari, Last=Nihalani	Full name: Mahak Kumari Nihalani
2	2	Same names	Length of full name = 21
3	3	Same names	Copied full name: Mahak Kumari Nihalani
4	4	First=Mahak, Last=Nihalani	First name and last name are different
5	5	Same names	Nihalani Kumari Mahak
6	6	Same names	Vowels=9, Consonants=10, Digits=0, Spaces=2
7	7	Search word: Kumari	Word found in the full name
8	7	Search word: Sharma	Word not found in the full name
9	9	Invalid menu choice	Invalid choice. Please enter 1 to 8
10	8	Exit	Exiting program

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

```

#define MAX 100
#define FULL 305

void readName(const char prompt[], char name[]);
void combineNames(const char firstName[], const char middleName[], const char lastName[], char fullName[]);
void findLength(const char fullName[], size_t *length);
void copyName(const char source[], char destination[]);
void compareNames(const char firstName[], const char lastName[], int *comparisonResult);
void reverseNameOrder(const char firstName[], const char middleName[], const char lastName[], char reverseName[]);
void countCharacters(const char fullName[], int *vowels, int *consonants, int *digits, int *spaces);
void searchWord(const char fullName[], const char word[], int *searchResult);

int main(void)
{
    char firstName[MAX], middleName[MAX], lastName[MAX];
    char fullName[FULL] = "", copiedName[FULL], reverseName[FULL], word[MAX];
    int choice, comparisonResult, vowels, consonants, digits, spaces, searchResult;
    size_t length;

    readName("Enter first name: ", firstName);
    readName("Enter middle name: ", middleName);
    readName("Enter last name: ", lastName);
    combineNames(firstName, middleName, lastName, fullName);

    do {
        printf("\n--- STUDENT NAME OPERATIONS MENU ---\n");
        printf("1. Combine and display full name\n");
        printf("2. Find length of full name\n");
        printf("3. Copy full name\n");
        printf("4. Compare first name and last name\n");
        printf("5. Display name in reverse order\n");
        printf("6. Count vowels, consonants, digits and spaces\n");
        printf("7. Search a word in the full name\n");
        printf("8. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice) {
            case 1:
                combineNames(firstName, middleName, lastName, fullName);
                printf("Full name: %s\n", fullName);
                break;
            case 2:
                findLength(fullName, &length);
                printf("Length of full name = %zu\n", length);
                break;
            case 3:
                copyName(fullName, copiedName);
                printf("Copied full name: %s\n", copiedName);
                break;
            case 4:
                compareNames(firstName, lastName, &comparisonResult);
                if (comparisonResult == 0)
                    printf("First name and last name are equal.\n");
                else
                    printf("First name and last name are different.\n");
                break;
            case 5:
                reverseNameOrder(firstName, middleName, lastName, reverseName);
                printf("Name in reverse order: %s\n", reverseName);

```

```

        break;
    case 6:
        countCharacters(fullName, &vowels, &consonants, &digits, &spaces);
        printf("Vowels = %d\n", vowels);
        printf("Consonants = %d\n", consonants);
        printf("Digits = %d\n", digits);
        printf("Spaces = %d\n", spaces);
        break;
    case 7:
        readName("Enter word to search: ", word);
        searchWord(fullName, word, &searchResult);
        if (searchResult)
            printf("Word found in the full name.\n");
        else
            printf("Word not found in the full name.\n");
        break;
    case 8:
        printf("Exiting program.\n");
        break;
    default:
        printf("Invalid choice. Please enter 1 to 8.\n");
    }
} while (choice != 8);

return 0;
}

void readName(const char prompt[], char name[])
{
    size_t len;
    printf("%s", prompt);
    fgets(name, MAX, stdin);
    len = strlen(name);
    if (len > 0 && name[len - 1] == '\n')
        name[len - 1] = '\0';
}

void combineNames(const char firstName[], const char middleName[], const char lastName[], char fullName[])
{
    strcpy(fullName, firstName);
    if (middleName[0] != '\0') {
        strcat(fullName, " ");
        strcat(fullName, middleName);
    }
    if (lastName[0] != '\0') {
        strcat(fullName, " ");
        strcat(fullName, lastName);
    }
}

void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}

void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

void compareNames(const char firstName[], const char lastName[], int *comparisonResult)
{

```

```

    *comparisonResult = strcmp(firstName, lastName);
}

void reverseNameOrder(const char firstName[], const char middleName[], const char lastName[], char reverseName[])
{
    strcpy(reverseName, lastName);
    if (middleName[0] != '\0') {
        strcat(reverseName, " ");
        strcat(reverseName, middleName);
    }
    if (firstName[0] != '\0') {
        strcat(reverseName, " ");
        strcat(reverseName, firstName);
    }
}

void countCharacters(const char fullName[], int *vowels, int *consonants, int *digits, int *spaces)
{
    size_t i;
    char ch;
    *vowels = *consonants = *digits = *spaces = 0;

    for (i = 0; i < strlen(fullName); i++) {
        ch = (char)tolower((unsigned char)fullName[i]);
        if (isalpha((unsigned char)fullName[i])) {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                (*vowels)++;
            else
                (*consonants)++;
        } else if (isdigit((unsigned char)fullName[i])) {
            (*digits)++;
        } else if (isspace((unsigned char)fullName[i])) {
            (*spaces)++;
        }
    }
}

void searchWord(const char fullName[], const char word[], int *searchResult)
{
    *searchResult = (strstr(fullName, word) != NULL);
}

```

Compilation and Execution

```

Gcc hdoc40.c -o hdoc40
./hdoc40

```

Compilation



```

(kali@kali)-[~/Desktop]
$ vi hdoc40.c

(kali@kali)-[~/Desktop]
$ gcc hdoc40.c -o hdoc40

```

Execution

```
(kali㉿kali)-[~/Desktop]
$ ./hdoc40
Enter first name: Mahak
Enter middle name: Kumari
Enter last name: Nihalani
```

Test Case 1:

```
8. Exit
Enter your choice: 1
Full name: Mahak Kumari Nihalani
```

Test Case 2:

```
8. Exit
Enter your choice: 2
Length of full name = 21
```

Test Case 3:

```
8. Exit
Enter your choice: 3
Copied full name: Mahak Kumari Nihalani
```

Test Case 4:

```
8. Exit
Enter your choice: 4
First name and last name are different.
```

Test Case 5:

```
8. Exit
Enter your choice: 5
Name in reverse order: Nihalani Kumari Mahak
```

Test Case 6:

```
8. Exit
Enter your choice: 6
Vowels = 9
Consonants = 10
Digits = 0
Spaces = 2
```

Test Case 7:

```
8. Exit
Enter your choice: 7
Enter word to search: Kumari
Word found in the full name.
```

Test Case 8:

```
8. EXIT
Enter your choice: 7
Enter word to search: Sharma
Word not found in the full name.
```

Test Case 9:

```
8. EXIT
Enter your choice: 9
Invalid choice. Please enter 1 to 8.
```

Test Case 10:

```
8. EXIT
Enter your choice: 8
Exiting program.
```